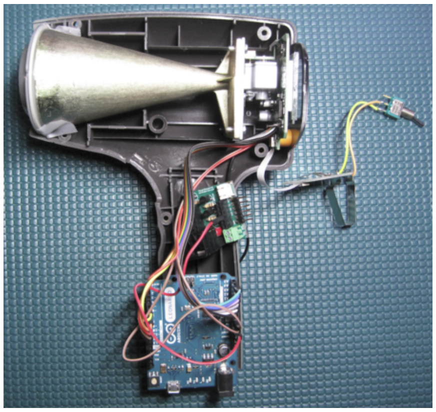
Arduino Radar Gun

From Kevin Darrah Wiki

...Back to Projects:



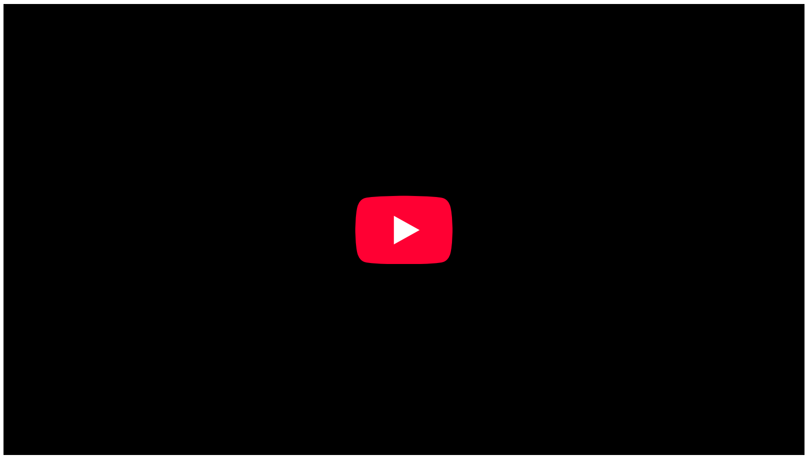
Contents

Arduino Radar Gun

Introduction

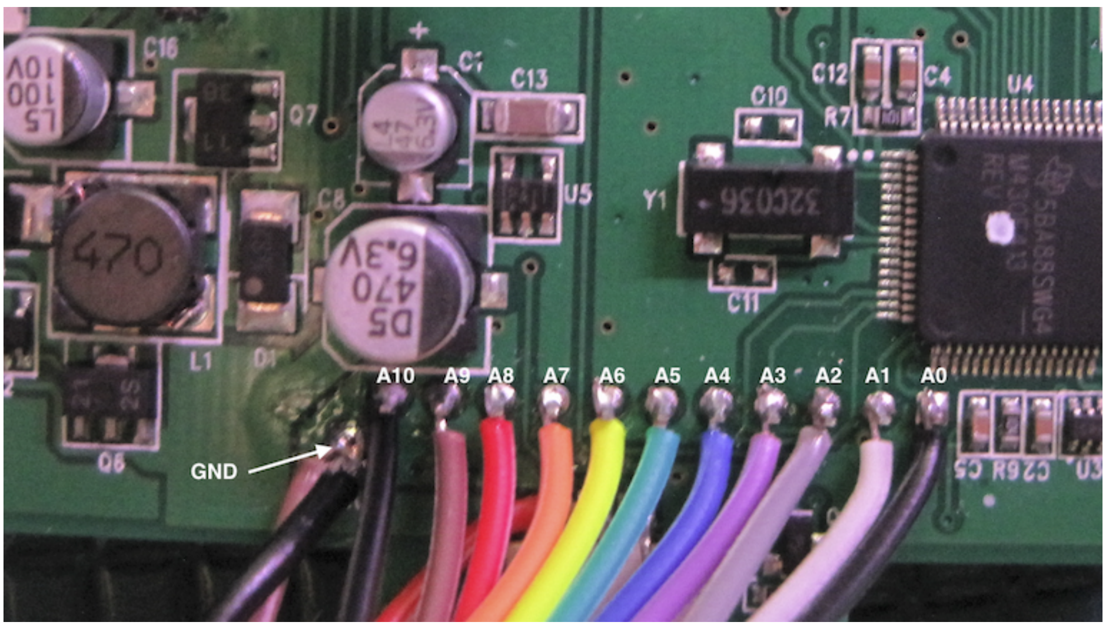
This is a long term project, so this page will be updated as the project continues. The goal here was to take control of an off-the-shelf radar gun. The gun used for the project was a "Bushnell Velocity Speed Gun" and was purchased on Amazon (https://www.amazon.com/gp/product/B0002X7V1Q/ref=oh_aui_detailpage_o04_s00?ie=UTF8&psc=1). An Arduino Leonardo (https://www.arduino.cc/en/Main/ArduinoBoardLeonardo) is being used due to it having x12 Analog Inputs. Since the Leonardo is based on the ATMEGA32U4 (http://www.atmel.com/Images/Atmel-7766-8-bit-AVR-ATmega16U4-32U4_Datasheet.pdf), any of the "Arduino Clones" would work, like this one (https://www.adafruit.com/products/3228)

Phase 1 - Reading the Speed Data

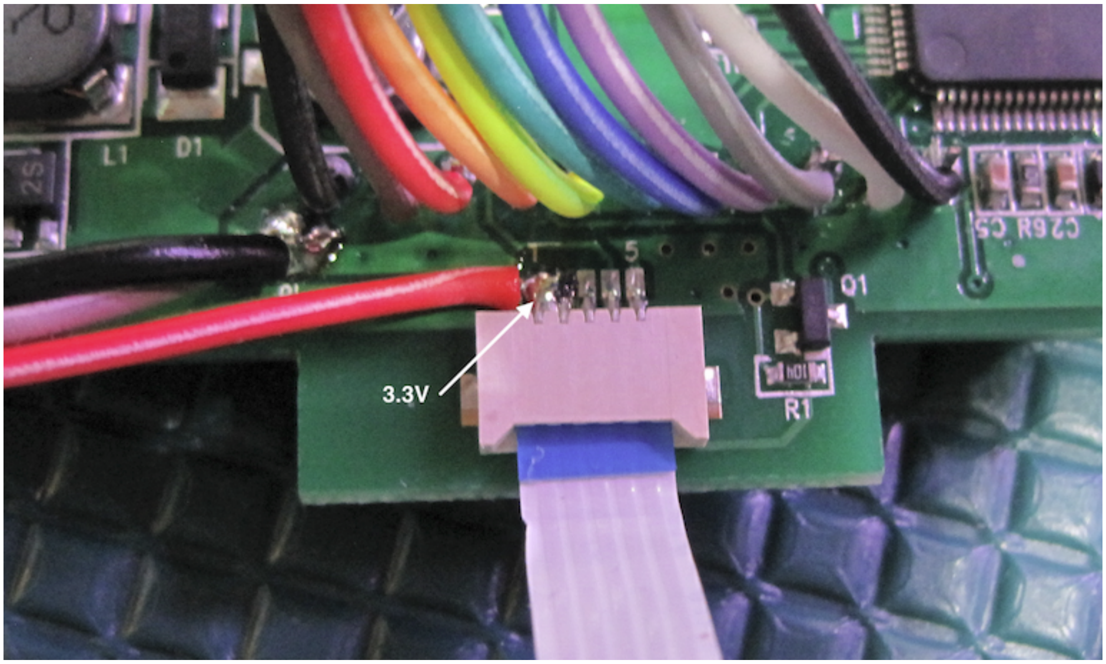


Connections

All x12 Analog Inputs are connected to the Numeric LCD pins on the gun. A0-A5 are the standard hookups to the Arduino, but A6-A10 go to: 4, 6, 8, 9, 10. Note that the ground connection is made where the battery clip wire terminates into the board



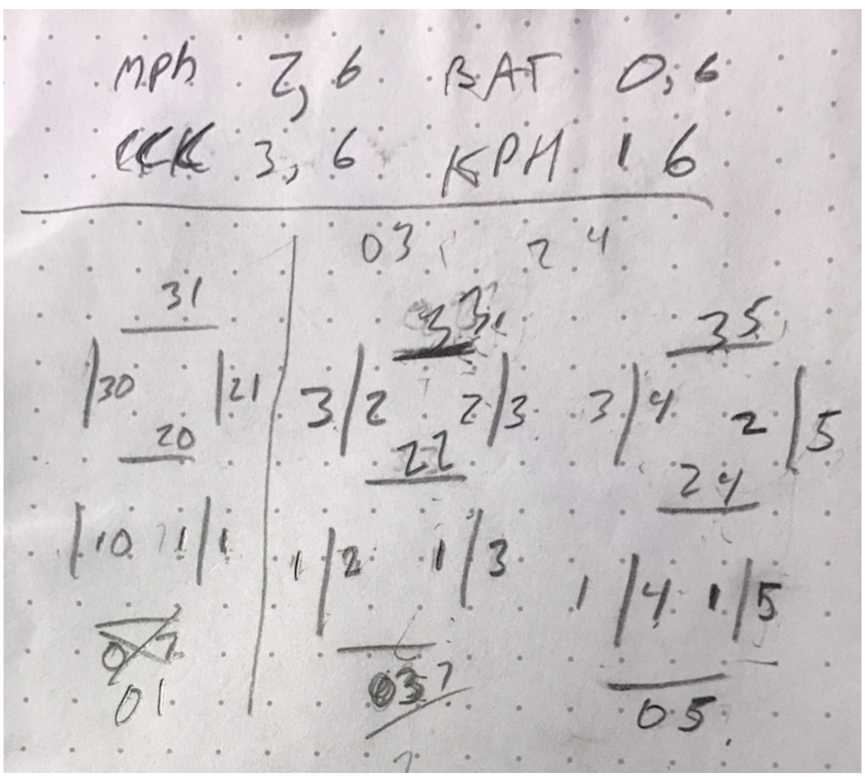
The trigger board cable brings up the positive 3V terminal from the battery stack, which is where I connected my 3.3V source wire. You can see that the two pins on the left are bridged together, which made the soldering much easier



Power to the Radar gun is supplied from the 5V USB, stepped down to 3.3V using a Linear Regulator "TC1262-3.3VAB". The Power Cheater was used for this. WARNING!!! The gun will pull over 200mA when you pull the trigger, so make sure the supply can handle that load current.

LCD Segment Mapping

Here's a shot of my notes when I was mapping the segments - I also took note of the misc characters like low battery, mph, kph, and the little doppler icon that pops up when you pull the trigger.

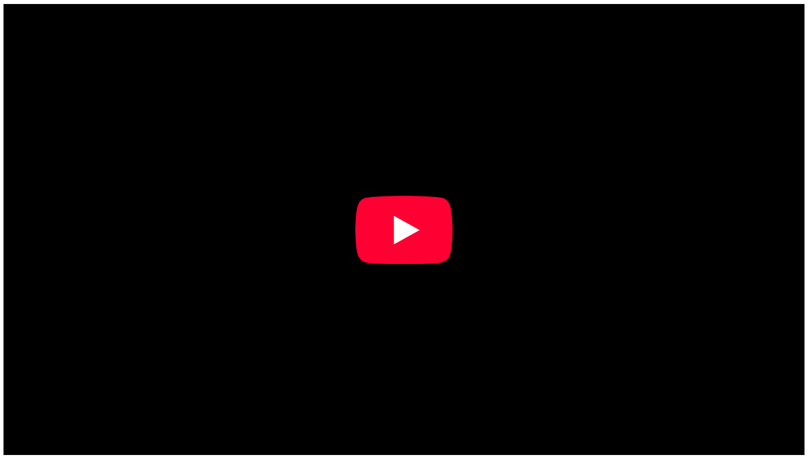


The Code

Download the Test Code (http://kevindarrah.com/download/arduino_code/RADARGUN_testRead.ino)

Radar Board

The radar board was developed so that a tiny board with the Arduino can be soldered down directly to the Radar Gun. The Arduino project can do anything you want from there. In the demo code below, it simply powers up the gun, triggers the radar gun to start scanning, then just outputs the speed over the USB interface. I also added in pads for a UART connection, so the data can be streamed out there as well. If the radar gun project is of no interest to you, this board makes a pretty slick little breakout for the ATMEGA32U4



Schematic